

DIRO Iframe Integration guide

The idea here is to provide a way to integrate the DIRO Verification process on any website with minimal amount of code.

Getting started

Using an **iframe** is the easiest method to integrate the DIRO verification process on any website.

Basically you get an iframe url, that can simply be used inside an iframe tag on any website and customize the iframe window's size accordingly and that's pretty much it, We take care of the rest of the inner working details and code inside the iframe.

DIRO initial deployment

1. Finalize the FAQ text and button placements
2. Finalize unique client Trackid (preferably email) to be used as reference to submissions
3. Host the HTML page at DIRO infra (or client) - client.diro.io
4. Following domains should be whitelisted by Client info security team to prevent getting the URL becoming marked as phishing website in tools like cloudflare. (must)

Domains to be whitelisted

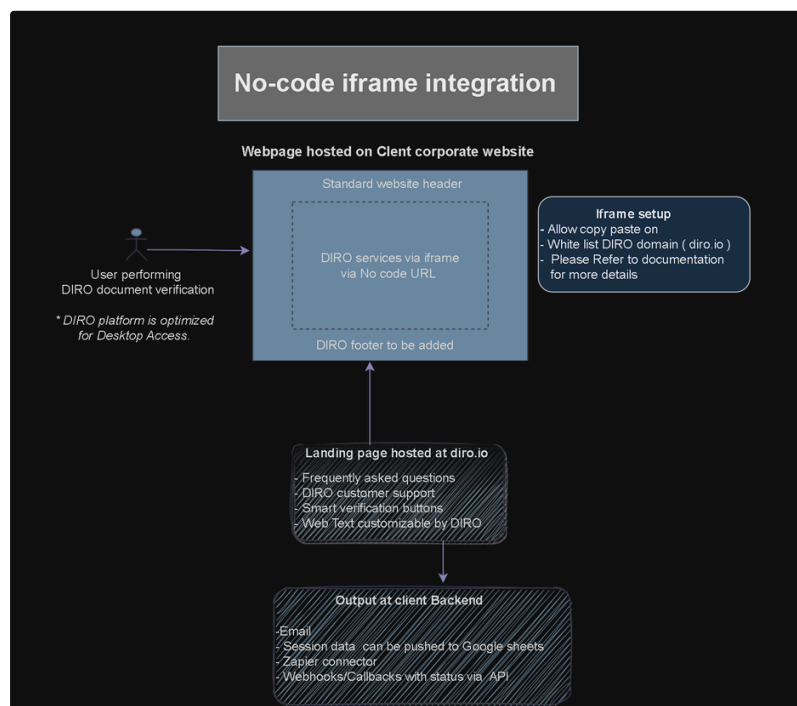
<https://geoip-js.com/>

*.azurefd.net

URL	USES
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/google.apis.css.css	google font
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/font-awesome.css	Font awesome. It is designed to be used with inline elements . The <code><i></code> and <code></code> elements are widely used for icons.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/bootstrap.css	Bootstrap css file.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/bootstrap.js	Bootstrap js file.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/select2.css	Used for custom html element. Select2 was designed to be a replacement for the standard <code><select></code> box that is displayed by the browser.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/jquery.js	custom.js 1 jQuery simplifies the HTML markup, document traversal, event handling, animating, and AJAX

	interactions for rapid web development.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/popper.js	custom.js 2 Popper.js is a positioning engine, its purpose is to calculate the position of an element to make it possible to position it near a given reference element.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/bundle.js	Custom.js 3 A JavaScript bundler can be used when your project becomes too large for a single file or when you're working with libraries that have multiple dependencies.
https://diro-cloudflare-cdn-gkffbhd8e0a2g7cs.z03.azurefd.net/cdn-assets/captureconsole.js	Custom.js 4 capture-console is a small Node.js library built to help when capturing log output via process.stdout and process.stderr

Client deployment on website



To integrate DIRO Iframe into your website, follow these simple steps:

1. Depending upon your website's structure you will have to create an iframe tag and paste the provided iframe url as the source of the iframe.

2. For example if the iframe url is: "https://diro-iframe.com"

then depending upon your website's structure the iframe code should look something like this:

```
<iframe src="https://diro-iframe.com" allow="clipboard-read *; clipboard-write *" frameborder="0"></iframe>
```

3. **Please add the client tracking id in the end if the landing page is already behind a log in page to avoid double login for the end users.**

To track or uniquely identify each verification session, you can provide a tracking id (preferably an email) which you can pass a parameter in the iframe url.

- Please note you must enable copy-paste in the iframe, this is done by adding the following fields in the iframe tag `allow="clipboard-read *; clipboard-write *"`
- Change the additional iframe setting according your needs, for example `width="100%"; height="100%"; frameBorder="0";` and so on.
- Here is another example using React.js

```
1 import IFram from "react-iframe";
2
3 <IFram
4   url="https://diro-iframe.com"
5   width="100%"
6   height="100%"
7   frameBorder="0"
8   id="ifr"
9   display="block"
10  allow="clipboard-read *; clipboard-write *"
11 />
```

****Iframe should be in webpage element that allows it to expand to at least 90% of width and maximum available.***

height of visible portion of web page. if max width and height is not possible the iframe will not open properly.

Additional options

- Link the page to any existing menu option or page in the website
- Create a new orphan subdomain that is not indexed or linked on the website
 - Issue relevant subdomain
 - Host the HTML page (from step 1) as a new independent page.
 - It will show as diro-verification.client.com

Double header issue on redirection (Successful submission / on Exit)

If you are integrating the DIRO Iframe which includes a header and sidebar, additional code will be required on your part to enable redirection upon successful submission or exit.

The DIRO capture process sends a postMessage to the top window on successful submission and on exit, these messages can be captured by adding an event listener and using which we can redirect to our homepage.

Add the following script to your page.

```
1 <script>
2   window.addEventListener('message', receiveMessage, false);
3   function receiveMessage(event) {
4     var redirectUrl = "your_homepage_url"
5     if(event.data.progress_status === "Submitted"){
6       setTimeout(() => {
7         window.location.href = redirectUrl;
8       }, 4000);
9     }
10    if(event.data === "Terminate session from guac"){
11      window.location.href = redirectUrl;
12    }
13  }
```

```
14 </script>
```

In the above script you just have to provide `redirectUrl`, which is the homepage where you want to be redirected after successful submission or on exit.

Add DIRO footer

It is ideal to add DIRO footer containing the links to terms of use and privacy below the iframe for whitelisting and customer confidence.

footer example:

Learn more about DIRO's [bank account verification](#) and [address verification](#) solution.

[Terms & Conditions](#) [Privacy Policy](#)

html code for the footer:

```
1 <div>
2   <p>
3     Learn more about DIRO's
4     <a href="https://diro.io/products/bank-account-verification/" target="_blank">
5       bank account
6     </a>
7     and
8     <a href="https://diro.io/products/utility-bill-verification/" target="_blank">
9       address verification
10    </a>
11    solutions.
12  </p>
13  <p>
14    <a href="https://diro.io/term-condition/" target="_blank">
15      Terms of Use.
16    </a>
17    <a href="https://diro.io/privacy-policy/" target="_blank">
18      Privacy Policy.
19    </a>
20  </p>
21 </div>
```

Security measures

1. Leveraging CSP (Content Security Policy):

- Elements of CSP Directive that DIRO implements and recommends:

These can be implemented on case-to-case basis on client's request.

CSP directives can prevent from attacks like Clickjacking, Cross Frame Scripting, Iframe Phishing, XSS etc.

`Content-Security-Policy: frame-ancestors` will allow framing of its landing page only by client's domain, each client has a dedicated landing page.

`Content-Security-Policy: frame-src` will make your site not allowed to render other sites other than included in the configurations.

- **Strict Sandbox Configuration:** We utilize the `sandbox` attribute with a strict configuration to limit the iframe's capabilities within your web page. This restricts actions like popups, executing scripts from external sources, or accessing cookies outside the iframe's domain.
- **Targeted `allow` Permissions:** If absolutely necessary for functionality, we will grant specific and minimal permissions using the `allow` attribute. These permissions will be clearly documented and only address essential functionalities required by the DIRO verification process.
- **`referrerpolicy` for Enhanced Privacy:** We set the `referrerpolicy` attribute to a policy like `no-referrer` or `origin` to prevent sending unnecessary information about your website when users interact with links within the iframe.

2. Secure Communication Channels:

- **Enforced HTTPS:** We enforce HTTPS for both our website and the DIRO document verification iframe. This ensures encrypted communication, protecting user data from eavesdropping and manipulation during transmission.

3. Mitigating Clickjacking and Cross- Frame Scripting Attacks:

- Cross-Frame Scripting (XFS) combines iframes with malicious JavaScript to steal data from users.
- `X-Frame-Options` Header: We leverage the `X-Frame-Options` header set by DIRO to control how our website can be embedded in iframes. This setting ensures that the verification process can only be initiated from our website, preventing clickjacking attacks.
- XFS attacks can be prevented by including `Content-Security-Policy: frame-ancestors` and `X-Frame-Options` and `X-XSS-Protection` headers in web server configuration.

4. Iframe Phishing:

- By default, content from an iframe can trigger top-level navigation. So, an attacker might leverage cross-site scripting (XSS) vulnerability on a web application to insert phishing code as an iframe to lead the user into a phishing website.
- `allow-top-navigation` is disabled to prevent against such type of attacks

5. Proactive Security Measures:

- **Input Validation and Sanitization:** We validate and sanitize any user input used to construct the iframe source URL. This prevents attackers from injecting malicious code through user input fields.
- **Monitoring for Suspicious Activity (Optional):** We may implement optional monitoring for suspicious iframe activity. This could involve logging iframe loads or utilizing CSP reporting features to identify potential violations.

Security policy headers:

```
1 headers:
2   - key: "X-XSS-Protection"
3     value: "1"
4   - key: "X-Content-Type-Options"
5     value: "nosniff"
6   - key: "Strict-Transport-Security"
7     value: "max-age=63072000; includeSubDomains; preload"
8   - key: "Content-Security-Policy"
9     value: "default-src 'self'; frame-src 'self' https://diro.io/ https://session.diro.io/; script-src 'self'
```